MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The line drawn from the eye of an observer to the point in the object viewed is termed as: [1]

(A) Horizontal line (B) Line of sight (C) Normal ray (D) Vertical plumb line

Q2. An observer looks up at an aeroplane flying in the sky. The angle between the horizontal line through the eye and the line of sight is called: [1]

(A) Angle of depression (B) Angle of elevation (C) Vertical angle (D) Reflected angle

Q3. An observer standing on top of a cliff looks down at a boat in the sea. The angle between the horizontal and line of sight is: [1]

(A) Angle of depression (B) Angle of elevation (C) Acute horizon (D) Zenith angle

Q4. If a vertical pole of height 18 m casts a shadow on the ground such that $\tan\theta = \frac{6}{7}$, then the length of the shadow is: [1]

(A) 21 m (B) 15.4 m (C) 24 m (D) 18 m

Q5. **Assertion (A):** If a ladder leaning against a wall makes an angle of $30°$ with the ground, then the distance of its foot from the wall is greater than the height reached on the wall.
Reason (R): For $\theta = 30°$, $\cot 30° = \sqrt{3} \approx 1.732 > 1$, meaning base $= \sqrt{3} \times$ height $>$ height. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. A tightrope walker climbs a rope of length 20 m attached from the ground to the top of a pole. If the rope makes an angle of $30°$ with the ground level, find the height of the pole. [2]

Q7. The angle of elevation of the top of a tower from a point on the ground is $30°$. If the observer moves 20 m towards the tower, the angle of elevation becomes $60°$. Find the height of the tower. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. As observed from the top of a 100 m high lighthouse from the sea level, the angles of depression of two ships on opposite sides of the lighthouse are $30°$ and $45°$. If the line joining the ships passes through the base of the lighthouse, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] [3]

Q9. A vertical flagstaff 6 m high stands on top of a pedestal. From a point on the ground, the angles of elevation of the bottom and top of the flagstaff are $30°$ and $60°$ respectively. Find the height of the pedestal. [3]

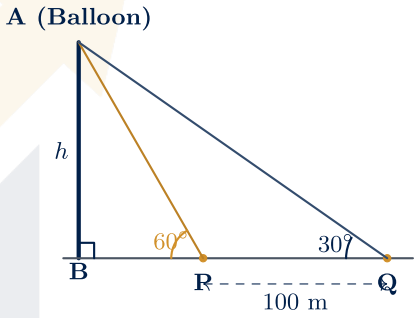
SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. A bird is sitting on the top of an 80 m high tree. From a point on the ground, the angle of elevation of the bird is 45°. The bird flies away horizontally at a constant height. After 2 seconds, the angle of elevation of the bird from the same observation point is 30°. Find the flying speed of the bird in m/s and km/h. [Take $\sqrt{3} = 1.732$] [5]

Q11. From the top of a tower 100 m high, an observer sees two cars on opposite sides of the tower on a straight horizontal road passing through the base. If their angles of depression are 45° and 60°, find the distance between the two cars. [Take $\sqrt{3} = 1.732$] [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Hot Air Balloon Festival Optical Tracking



At an international aerostat festival, a tethered research balloon ascends vertically into the atmosphere directly above a launch pad B.
Two optical tracking stations, P and Q, are stationed on level ground at a distance of 100 m apart along the same radial line from the launch pad. At a specific measurement instant, station P (nearer) observes the balloon (A) at an elevation of 60°, while station Q observes it at an elevation of 30°.
Based on the above mathematical scenario, answer the following questions:

(i) Express the horizontal ground distance from station P to the launch pad (BP) in terms of the altitude h. [1]

(ii) Express the horizontal ground distance from station Q to the launch pad (BQ) in terms of the altitude h. [2]

(iii) Calculate the exact vertical altitude h of the balloon. [Take $\sqrt{3} = 1.732$] [2]

— OR —

Find the numerical horizontal distance of tracking station P from the launch pad B. [2]

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★